

Simulating Wound Healing

Consider how wound healing can be described by the Porous–Fisher equation:

$$\frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left(\alpha \left(\frac{u}{K} \right)^m \frac{\partial u}{\partial x} \right) + \beta u \left(1 - \frac{u}{K} \right).$$

where $u(x, t)$ denotes the density of cells and $K > 0$ is the carrying capacity. Consider $0 \leq x \leq L$ and $t > 0$, subject to the following initial and boundary conditions:

$$u(x, 0) = f(x) = \begin{cases} 1, & x \leq \frac{L}{4} \text{ or } x \geq \frac{3L}{4}, \\ 0, & \frac{L}{4} < x < \frac{3L}{4}. \end{cases}$$

$$\frac{\partial u}{\partial x}(0, t) = 0, \quad \frac{\partial u}{\partial x}(L, t) = 0, \quad t > 0.$$

The initial condition represents a one-dimensional cross-section of a wound located on the interval

$$\frac{L}{4} < x < \frac{3L}{4},$$

where the cell population is initially zero.

1 Finite-volume spatial discretisation

We discretise the initial-boundary value problem in space using the finite volume method. Averaging is used to discretise the nonlinear diffusion term, and a non-uniform mesh is allowed with nodes

$$x = x_i, \quad i = 1, \dots, N,$$

where

$$x_1 = 0, \quad x_N = L, \quad N \in \mathbb{Z}^+.$$

Let u_i be the numerical approximation to $u(x_i, t)$. The spatial discretisation is expressed in the form

$$\frac{d\vec{u}}{dt} = \vec{G}(\vec{u}), \quad \vec{u}(0) = \vec{u}^{(0)}.$$

Here,

$$\vec{u} = (u_1, \dots, u_N)^T, \quad \vec{u}^{(0)} = (f(x_1), \dots, f(x_N))^T,$$

and

$$\vec{G}(\vec{u}) = (g_1(\vec{u}), \dots, g_N(\vec{u}))^T.$$

2 Discretisation final result

Define the nonlinear diffusivity and reaction term as

$$D(u_i) = \alpha \left(\frac{u_i}{K} \right)^m,$$

and

$$R(u_i) = \beta u_i \left(1 - \frac{u_i}{K} \right).$$

For a non-uniform mesh, define the node spacings

$$h_i = x_{i+1} - x_i, \quad i = 1, \dots, N-1.$$

The control volume lengths are

$$V_1 = \frac{h_1}{2}, \quad V_i = \frac{h_{i-1} + h_i}{2}, \quad V_N = \frac{h_{N-1}}{2}.$$

Using East and West notation, define the averaged diffusivities at the west and east faces as

$$D_{W,i} = \frac{D(u_{i-1}) + D(u_i)}{2}, \quad D_{E,i} = \frac{D(u_i) + D(u_{i+1})}{2}.$$

The spatial derivatives at the west and east faces are approximated by

$$\frac{\partial u}{\partial x}(w_i, t) \approx \frac{u_i - u_{i-1}}{h_{i-1}}, \quad \frac{\partial u}{\partial x}(e_i, t) \approx \frac{u_{i+1} - u_i}{h_i}.$$

Therefore,

$$g_i(\vec{u}) = \begin{cases} \frac{1}{V_1} \left[-D_{E,1} \frac{\partial u}{\partial x}(e_1, t) \right] + R(u_1), & i = 1, \\ \frac{1}{V_i} \left[D_{W,i} \frac{\partial u}{\partial x}(w_i, t) - D_{E,i} \frac{\partial u}{\partial x}(e_i, t) \right] + R(u_i), & i = 2, \dots, N-1, \\ \frac{1}{V_N} \left[D_{W,N} \frac{\partial u}{\partial x}(w_N, t) \right] + R(u_N), & i = N. \end{cases}$$